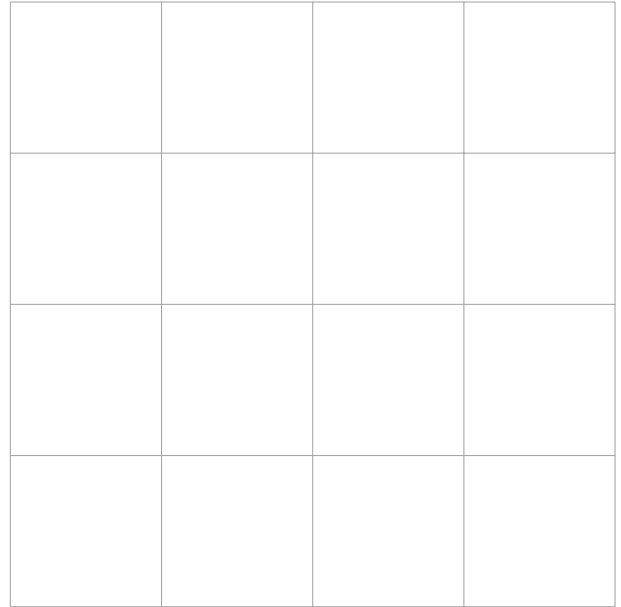
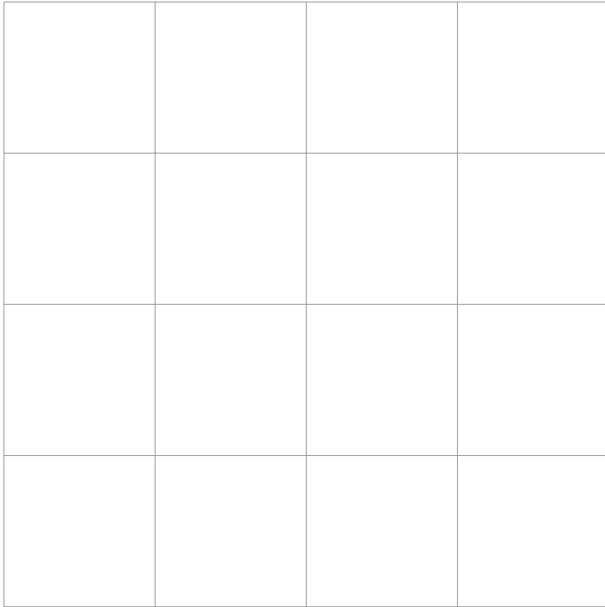


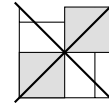
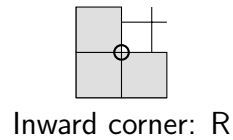
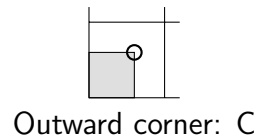
Problem 1: Fill a 4×4 square with eight R tiles and eight B tiles. Each room must be joined through tile sides. Make the shared fence between the rooms as short as possible. Count one unit for each tile side.



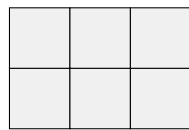
R boundary	B boundary	Shared fence

Problem 2: Keep the same 4×4 square. Try several divisions into two rooms. Find a rule connecting the two room-boundary lengths, the shared fence, and the outside boundary of the square.

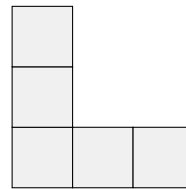
For these shapes, tiles must join through sides. At every grid corner, forbid the two-diagonal-tile pattern shown crossed out, even if those tiles connect elsewhere. Count corners on every boundary, including holes.



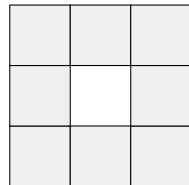
Problem 3: Build these shapes. Count their outward corners C, inward corners R, and holes. What rule connects the three counts?



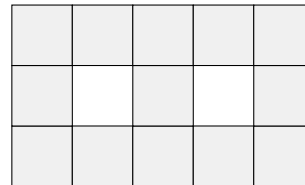
1



2



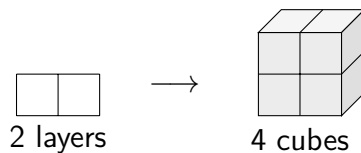
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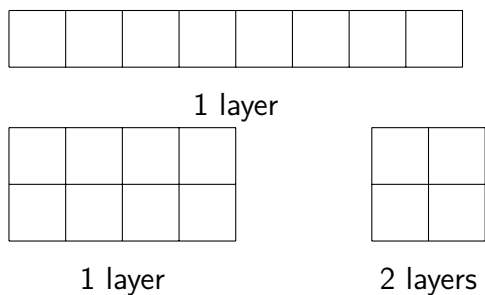
4

Shape	C	R	Holes
1			
2			
3			
4			

Each footprint square supports the stated number of cubes in a solid vertical stack. Faces where cubes touch are covered. Count all other faces, including the bottom.



Problem 4: Build each of these three buildings with eight cubes. Count every exposed square face. Which building has the least exposed surface?



Row of eight	2×4 , one layer	2×2 , two layers

Problem 5: Make a building with equal-height stacks on every square of a footprint. How can the footprint's area, its boundary length, and the number of layers tell you the exposed-face count? Find a rule that works for footprints with holes too.

Problem 6: Use a 2×2 footprint. Put stacks of heights 1, 2, 3, and 4 on its four squares, using each height once. Arrange them to make the fewest exposed faces, and then the most.

